FLEET PERFORMANCE & FUEL LOSS INVESTIGATION

Data Analyst / Business Intelligence Case Study

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Role: Data Analyst

Project Duration: July 2026

Domain: Operational Fleet & Fuel Analytics

1. Business Context, Problem & Objectives

The project sits within an operational fleet and fuel-monitoring context. The supplied practical assessment requires analysis of operational fleet and fuel data, anomaly identification, KPI calculation, business insights, fuel investigation and management reporting.

Business Problem

Fleet performance needed to be evaluated across distance, fuel use, fuel efficiency, idle time and route compliance, while fuel-event records needed to be screened for suspicious patterns.

Business Questions

- Which vehicles are least fuel efficient?
- Which drivers have unusually high idle times?
- Which vehicles show abnormal fuel consumption?
- Which vehicles have poor route compliance?
- What suspicious fuel-drop patterns appear in the event data?
- Which fuel drops occur during unusual hours or on remote routes?
- Which vehicles have repeated fuel-drop incidents?
- What actions can reduce fuel losses and improve operational control?

Project Objectives

- Validate Fleet_Data and Fuel_Events and document data-quality issues.
- Calculate fleet KPIs and vehicle/driver rankings.
- Investigate fuel efficiency, idle time, route compliance and fuel-drop anomalies.
- Translate analytical results into operational recommendations.
- Present the results in a management-oriented Power BI dashboard and executive summary.

Business Understanding → Data Validation → KPI Analysis → Business Insights → Fuel Investigation → Visualization → Power BI Dashboard → Executive Summary → Recommendations

2. Dataset Overview & Data Understanding

Two datasets were supplied: Fleet_Data and Fuel_Events. The workbook contains 804 Fleet_Data records across 9 columns and 303 Fuel_Events records across 6 columns. Fleet_Data spans 1 January 2026 to 31 March 2026.

Dataset	Records	Columns	Purpose
Fleet_Data	804	9	Daily fleet operational records
Fuel_Events	303	6	Fuel-event log for anomaly investigation

Fleet_Data Structure

Column	Type	Meaning
Vehicle_ID	Text	Vehicle identifier
Driver_Name	Text	Assigned driver name
Date	Date	Operational record date
Distance_KM	Numeric	Distance travelled
Fuel_Consumed_L	Numeric	Fuel consumed
Idle_Time_Hrs	Numeric	Idle time
Trips	Numeric	Trip count
Average_Speed_KMH	Numeric	Average speed
Route_Compliance_Pct	Numeric	Route compliance percentage

Fuel_Events Structure

Column	Type	Meaning
Vehicle_ID	Text	Vehicle associated with event
Timestamp	Date/time	Event timestamp
Event_Type	Text	Event category
Quantity_L	Numeric	Fuel quantity
Location	Text	Event location
HOUR	Derived	Hour extracted from timestamp

3. Data Validation & Cleaning

The supplied workbook documents validation checks for missing values, duplicates and suspicious records. Fleet_Data had five missing values in each of Fuel_Consumed_L, Idle_Time_Hrs and Route_Compliance_Pct; the workbook states these were replaced with the respective column mean. Six duplicate Fleet_Data records were removed. Fuel_Events had no reported missing or duplicate records.

Data Issue	Action	Reason / Result
5 missing Fuel_Consumed_L	Mean replacement	Enables KPI calculation
5 missing Idle_Time_Hrs	Mean replacement	Enables idle-time analysis
5 missing Route_Compliance_Pct	Mean replacement	Enables compliance analysis
6 duplicate Fleet_Data records	Removed	Prevents duplicated records distorting analysis
0 Fuel_Events missing values	No action	None reported
0 Fuel_Events duplicates	No action	None reported

Suspicious Fleet Records

Vehicle	Issue	Evidence
VH-001	High fuel consumption	250 KM but 120 L consumed
VH-002	High idle time	12 hours idle; average speed 20 KM/H
VH-003	Unrealistic efficiency	400 KM using only 15 L
VH-004	Poor route compliance	40% compliance

4. Data Transformation & KPI Development

The supplied workbook uses aggregation and calculation outputs to convert operational records into management metrics. Average efficiency is calculated as total distance divided by total fuel.

KPI	Calculation	Actual Value	Business Meaning
Total Distance	SUM Distance_KM	212,316 KM	Fleet activity
Total Fuel	SUM Fuel_Consumed_L	52,569.7 L	Reported fuel use
Average Efficiency	Distance ÷ Fuel	4.04 KM/L	Fuel productivity
Average Idle Time	AVERAGE Idle_Time_Hrs	2.36 hrs	Idle behavior
Average Route Compliance	AVERAGE Route_Compliance_Pct	92.54%	Route adherence

Vehicle Rankings

Ranking	Vehicles	Metric
Top 5 by distance	VH-018, VH-011, VH-003, VH-020, VH-005	14,685.8; 14,013.8; 13,386.5; 12,803.5; 12,650.3 KM
Top 5 by fuel	VH-018, VH-011, VH-020, VH-005, VH-003	3,629.2; 3,334.1; 3,323.6; 3,237.2; 3,214.7 L
Bottom 5 efficiency	VH-009, VH-001, VH-020, VH-005, VH-012	3.71; 3.82; 3.85; 3.91; 3.95 KM/L

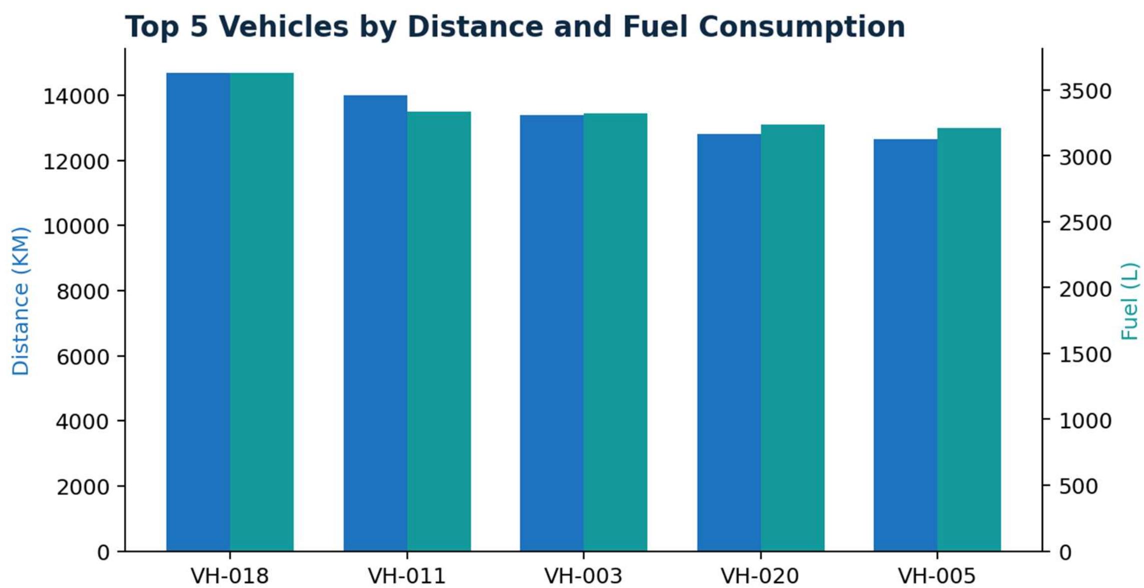

Figure 1. Top-five vehicles by distance and fuel consumption.

5. Exploratory Data Analysis

Fuel Consumption Trend

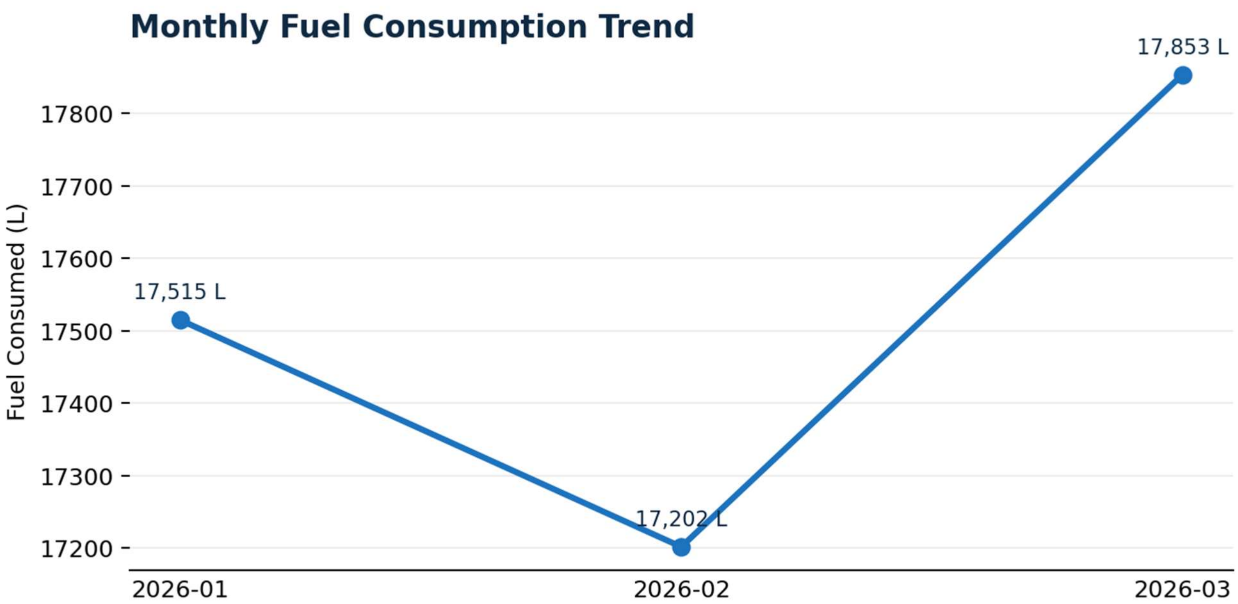

Figure 2. Monthly fuel consumption trend.

Month	Distance (KM)	Fuel (L)	Avg Idle	Avg Compliance
1	71,482.9	17,514.8	2.42	92.85%
2	69,188.2	17,201.6	2.42	92.24%
3	71,644.9	17,853.3	2.24	92.52%

February records the lowest monthly fuel consumption at 17,201.6 L, while March is highest at 17,853.3 L. January is 17,514.8 L. The supplied materials do not establish a causal reason for this movement.

Vehicle Fuel Efficiency

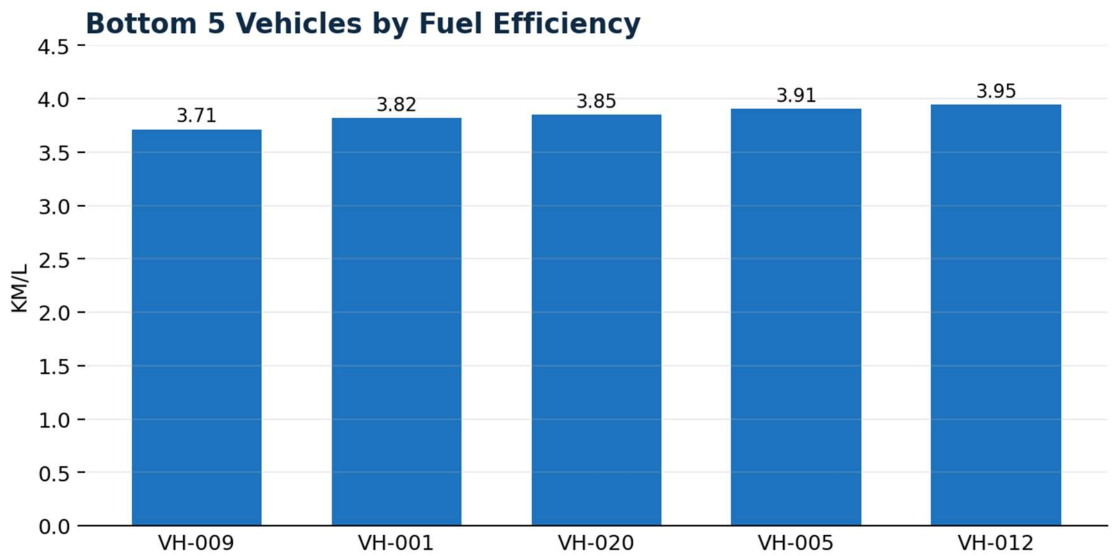

Figure 3. Bottom-five vehicles by fuel efficiency.

The five lowest-efficiency vehicles are VH-009 (3.71 KM/L), VH-001 (3.82), VH-020 (3.85), VH-005 (3.91) and VH-012 (3.95).

Driver Idle Time

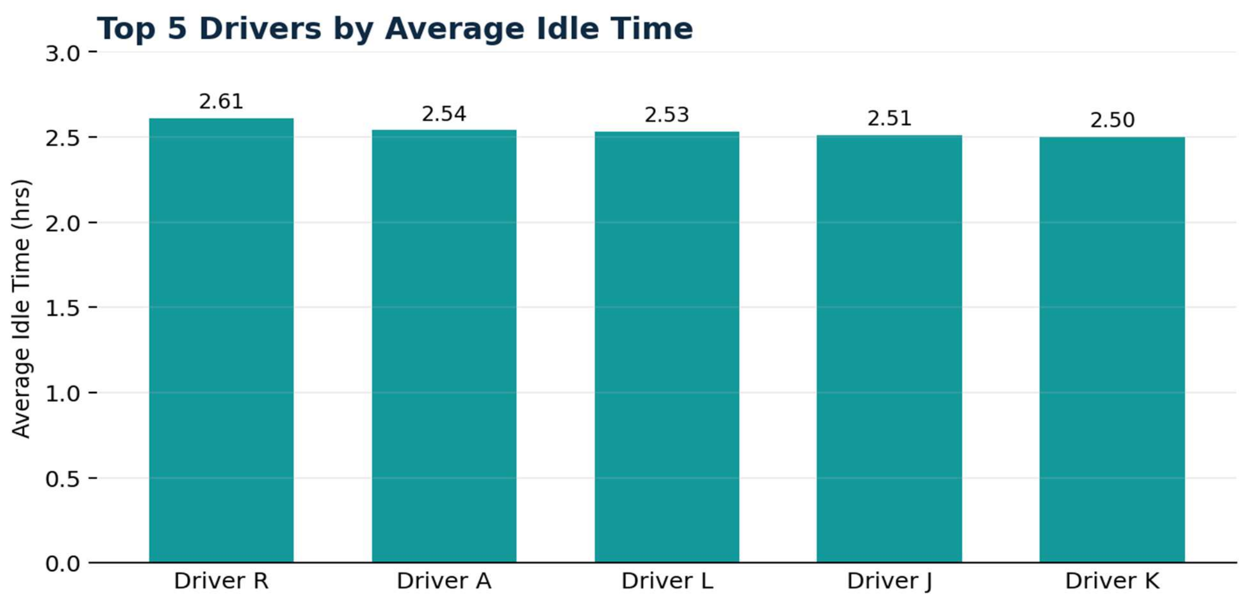

Figure 4. Top-five drivers by average idle time.

Driver R has the highest average idle time at 2.61 hours, followed by Driver A (2.54), Driver L (2.53), Driver J (2.51) and Driver K (2.50).

6. Route Compliance

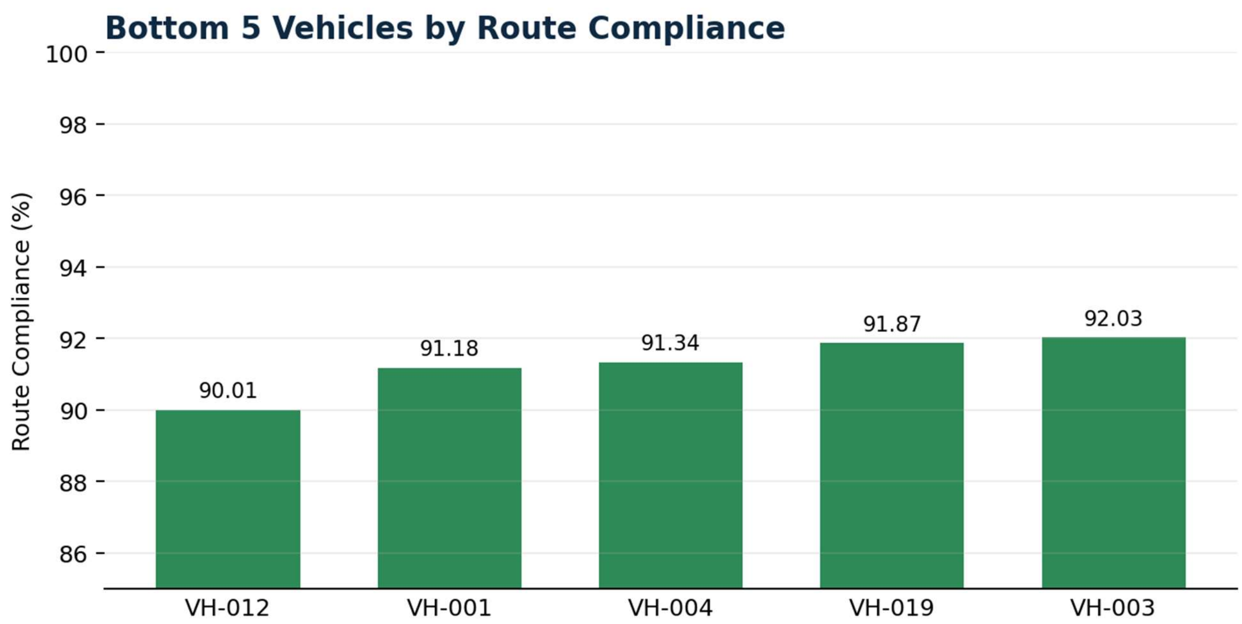

Figure 5. Bottom-five vehicles by route compliance.

Vehicle	Compliance
VH-012	90.01%
VH-001	91.18%
VH-004	91.34%
VH-019	91.87%
VH-003	92.03%

Fleet average route compliance is 92.54%. The executive summary highlights route deviation as a fuel and safety risk. The supplied recommendation uses below 90% as a control threshold; the bottom-five ranking is therefore not identical to the below-90% group.

7. Fuel-Drop Investigation & Risk Screening

Investigation KPI	Value
Total fuel-drop events	96
Drops above 40 L	38
Remote-route drops	3
Night-time drops (10 PM–6 AM)	39

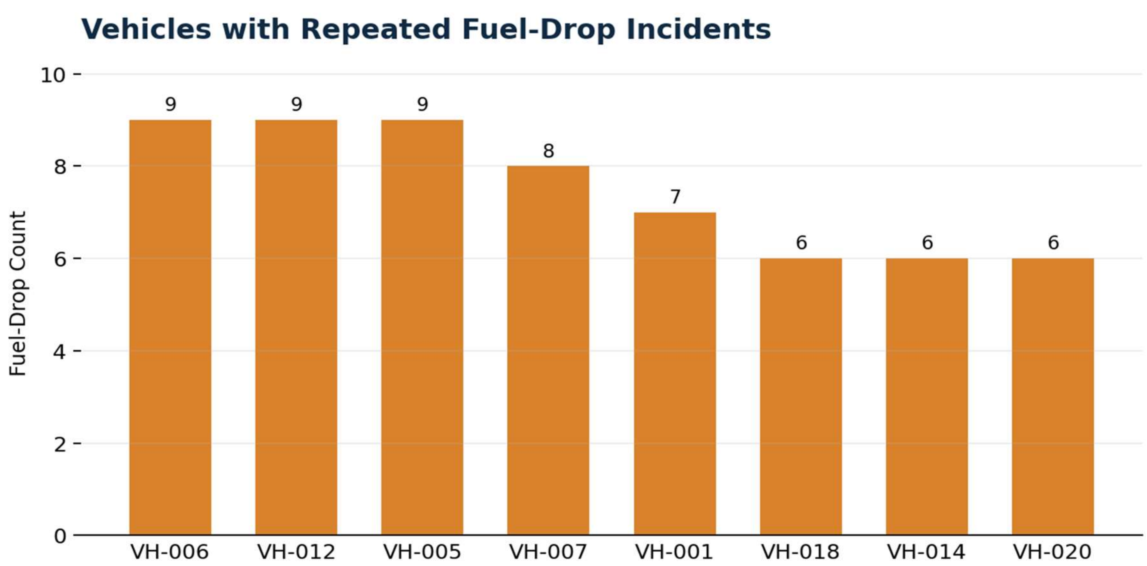

Figure 6. Vehicles with repeated fuel-drop incidents.

VH-006, VH-012 and VH-005 each recorded 9 drops, followed by VH-007 with 8 and VH-001 with 7.

Highest-Risk Remote-Route Cases

Vehicle	Timestamp	Drop (L)	Location
VH-001	2026-02-15 02:15	48.5	Remote Route
VH-001	2026-02-20 01:40	52.1	Remote Route
VH-007	2026-03-05 03:10	44.3	Remote Route

Other large unusual-hour drops recorded in the workbook include VH-016 (56.8 L at 04:47), VH-014 (55.3 L at 23:38), VH-005 (55.1 L at 03:47), VH-007 (53.2 L at 05:26), VH-019 (48.1 L at 05:12) and VH-003 (45.9 L at 04:23).

8. Power BI Dashboard

The supplied Power BI dashboard provides the final management-facing view. It contains a date slicer, Vehicle ID and Driver Name filters, four KPI cards, a monthly fuel-consumption trend, a top-five driver idle-time chart, a top-five vehicle efficiency chart and a top-five vehicle route-compliance chart.

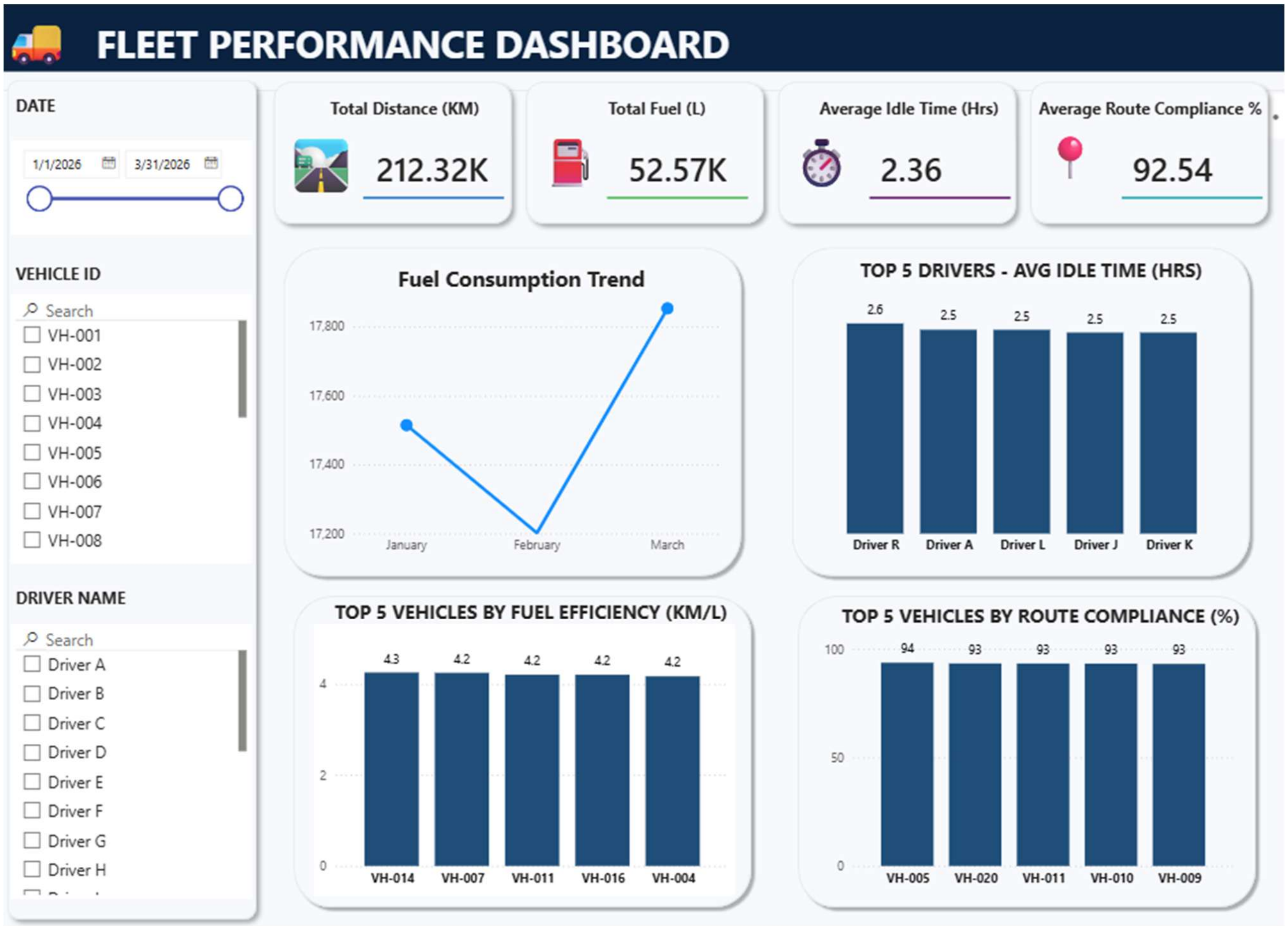

Figure 7. Supplied Power BI Fleet Performance Dashboard.

Dashboard User Journey

The dashboard begins with headline KPIs for total distance, total fuel, average idle time and route compliance. The trend section then shows monthly fuel movement, while ranking visuals answer focused questions about drivers and vehicles. Date, vehicle and driver slicers support targeted exploration.

The screenshot does not provide evidence of drill-through pages, bookmarks or custom tooltip pages; these are Not Provided.

9. Key Findings & Business Recommendations

Finding	Evidence	Action
Low-efficiency vehicles	VH-009, VH-001, VH-020, VH-005 and VH-012 are the bottom five at 3.71–3.95 KM/L.	Prioritize efficiency diagnostics and maintenance.
High-idle drivers	Driver R, A, L, J and K range from 2.50–2.61 idle hours.	Use idle monitoring and eco-driving / idle-reduction training.
Night and remote fuel-drop risk	96 total drops; 38 above 40 L; 39 at night; 3 on remote routes.	Investigate the three remote-route cases first.
Repeated fuel-drop vehicles	VH-006, VH-012 and VH-005 each have 9 drops; VH-007 has 8; VH-001 has 7.	Consider targeted fuel controls on high-count vehicles.
Route-compliance monitoring	Fleet average is 92.54%; bottom-five range from 90.01% to 92.03%.	Use route audits and geo-fence enforcement.

Recommended Control Actions

- Immediately investigate the three remote-route fuel drops involving VH-001 twice and VH-007 once using GPS data, trip logs and driver interviews.
- Flag night-time fuel drops between 10 PM and 6 AM above 30 L for security review and mandatory driver interview.
- Install secondary fuel-level sensors or lockable fuel caps on vehicles with the highest drop counts.
- Cross-match fuel-drop timestamps with GPS and trip logs and set alerts for drops above 25 L outside authorized fuel stations.
- Provide eco-driving and idle-reduction training, service low-efficiency vehicles and reinforce route discipline

10. Business Problem Solved & Conclusion

Before Analysis

The supplied operational data required validation, KPI calculation, anomaly screening and management presentation. The analysis consolidated these dimensions into one decision-oriented view.

Analysis Performed

The project validated data quality, handled documented missing values and duplicates, identified suspicious records, calculated fleet KPIs, ranked vehicles and drivers, analyzed monthly fuel consumption, screened fuel-drop events and consolidated results in Power BI and an executive summary.

Conclusion

The analysis shows 212,316 KM of reported distance, 52,569.7 L of reported fuel consumption, 4.04 KM/L average efficiency, 2.36 hours average idle time and 92.54% average route compliance. At the event level, 96 fuel drops were identified, including 38 above 40 L, 39 during the 10 PM–6 AM window and three remote-route cases. The strongest management priorities are improving operational efficiency and strengthening controls around large, repeated, night-time and remote-route fuel events.